

1.5.2 Differentiation

Here is the derivative of x^n with respect to x .

```
In[1]:= D[ x^n , x ]
Out[1]= n x-1 + n
```

Mathematica knows the derivatives of all the standard mathematical functions.

```
In[2]:= D[ ArcTan[x] , x ]
Out[2]=  $\frac{1}{1 + x^2}$ 
```

This differentiates three times with respect to x .

```
In[3]:= D[ x^n , {x, 3} ]
Out[3]= n x-3 + n (-2 + n) (-1 + n)
```

The function $D[x^n, x]$ really gives a *partial* derivative, in which n is assumed not to depend on x . Mathematica has another function, called Dt , which finds *total* derivatives, in which all variables are assumed to be related. In mathematical notation, $D[f, x]$ is like $\frac{\partial f}{\partial x}$, while $Dt[f, x]$ is like $\frac{df}{dx}$. You can think of Dt as standing for “derivative total”.

Dt gives a *total derivative*, which assumes that n can depend on x . $Dt[n, x]$ stands for $\frac{dn}{dx}$.

```
In[4]:= Dt[ x^n , x ]
Out[4]= n x-1 + n + x^n Dt[n, x] Log[x]
```

This gives the total differential $d(x^n)$. $Dt[x]$ is the differential dx .

```
In[5]:= Dt[ x^n ]
Out[5]= n x-1 + n Dt[x] + x^n Dt[n] Log[x]
```

$D[f, x]$	partial derivative $\frac{\partial}{\partial x} f$
$D[f, x_1, x_2, \dots]$	multiple derivative $\frac{\partial}{\partial x_1} \frac{\partial}{\partial x_2} \dots f$
$D[f, \{x, n\}]$	repeated derivative $\frac{\partial^n f}{\partial x^n}$
$Dt[f]$	total derivative df
$Dt[f, x]$	total derivative $\frac{d}{dx} f$

Some differentiation functions.

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As well as treating variables like x symbolically, you can also treat functions in *Mathematica* symbolically. Thus, for example, you can find formulae for derivatives of $f[x]$, without specifying any explicit form for the function f .

Mathematica does not know how to differentiate f , so it gives you back a symbolic result in terms of f' .

```
In[6]:= D[ f[x], x ]
Out[6]= f'[x]
```

Mathematica uses the chain rule to simplify derivatives.

```
In[7]:= D[ 2 x f[x^2], x ]
Out[7]= 2 f[x^2] + 4 x^2 f'[x^2]
```